

# **Automated systems**

## **Automated and emerging technologies**

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# Sensors, microprocessors and actuators

An automated system is a combination of software and hardware (for example, sensors, microprocessors and actuators) that is designed and programmed to work automatically without the need of any human intervention. However, such systems often involve human monitoring.

- Sensors are input devices that take readings from their surroundings and send this data to a microprocessor or computer. If the data is analogue, it is first converted into a digital format by an analogue-digital converter (ADC).
- The microprocessor will process the data and take the necessary action based on programming.
- This will involve some form of output, usually involving signals sent to actuators to control motors, wheels, solenoids, and so on.

# Advantages and disadvantages of automated systems

A number of examples will be used to show the advantages and disadvantages of using automated systems. This list is by no means exhaustive, and simply intends to show the role of sensors, microprocessors (or computers) and actuators in the following application areas:

- industrial
- transport
- agriculture
- weather
- gaming
- lighting
- science.

## Industrial applications

In recent years, the focus on increased automation has led to improved quality and flexibility. For example, in the manufacture of car engines, when done manually, the installation of pistons into the engine had an error-rate of ~1.5%; with automated systems, the error-rate has fallen to 0.00001%.

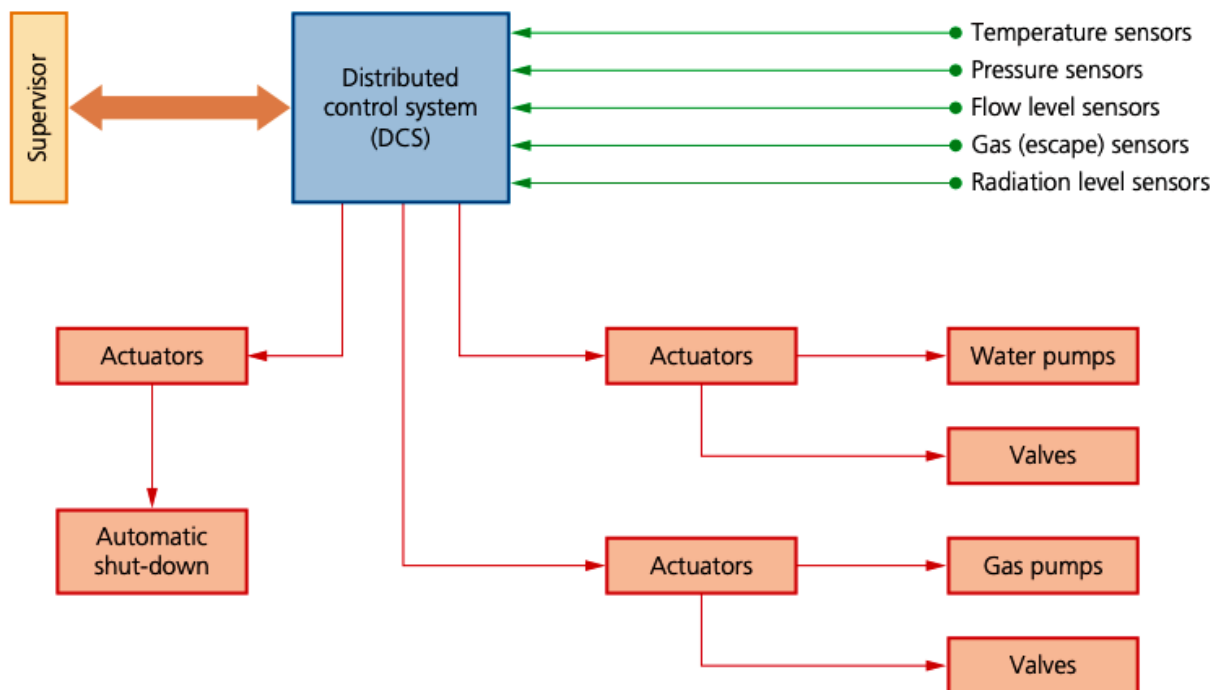
We will now consider two very different industrial applications.

# Example 1

## A nuclear power station

A key use of automated systems is in the control and monitoring of a nuclear power station. This is a good example, since automation gives increased safety both in the process itself and to the workforce. At the centre of the system is a distributed control system (DCS). DCS is essentially a powerful computer that has been programmed to monitor and control the whole process with no human interaction required:

Data from a number of sensors is sent to a DCS (computer) – if the data is analogue, it must first be converted into digital format using an ADC. The DCS will have access to a large database containing operational data and parameters. If any action needs to be taken, then signals will be sent to the appropriate actuators to operate pumps, valves or even an emergency shutdown system. The key here is that the system is fully automated. A human operator (the supervisor) will sit in a remote control room where a schematic of the process will show on a large screen. While the process is fully automatic, the supervisor can still override the DCS and shut down the process.



The main advantages of this automated system are:

- much faster than a human operator to take any necessary action
- much safer (an automated system is more likely to make timely interventions than a human; it also keeps humans away from a dangerous environment)
- the process is more likely to run under optimum conditions since any small changes needed can be identified very quickly and action taken
- in the long run, it is less expensive (an automatic system replaces most of the workforce who would need to monitor the process 24 hours a day).

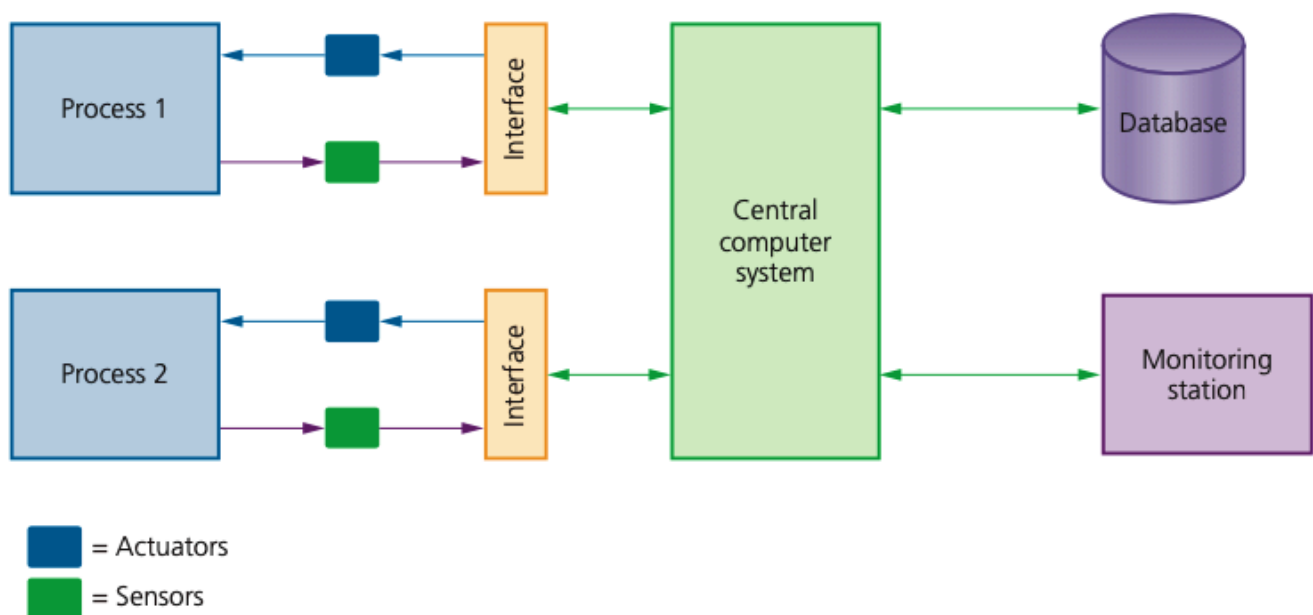
The main disadvantages of this automated system are:

- expensive to set up in the first place and needs considerable testing
- always possible for a set of conditions to occur that were never considered during testing which could have safety implications (hence the need for a supervisor)
- any computerised system is subject to cyberattacks no matter how good the system (one way round this is to have no external links to the DCS; although the weak link could potentially be the connection to the supervisor)
- automated systems always need enhanced maintenance which can be expensive.

# Example 2

## Manufacture of paracetamol

This automated system also depends on sensors, a computer, actuators and software. Process 1 is the manufacture of the paracetamol. Process 2 is the making of the solid tablets. Both processes are monitored by a number of sensors that send their data back to a central computer. The computer consults its database to ensure both processes are operating within correct parameters. Any necessary action is taken by the computer, sending signals to the appropriate actuator to operate pumps, valves, heaters, stirrers or pistons to ensure both processes can operate without any human intervention. Again, this system uses a remote monitoring station manned by an operator. The system is fully automated, but the operator can override the central computer system if necessary.



The main advantages of this automated system are:

- much faster than a human operator to take any necessary action
- much safer (an automated system is more likely to make timely interventions than a human if necessary; it also keeps humans away from a potentially dangerous environment)
- the process is more likely to run under optimum conditions since any small changes needed can be identified very quickly and action taken
- in the long run, it is less expensive (an automatic system replaces most of the workforce who would need to monitor the process 24 hours a day)
- more efficient use of materials
- higher productivity
- more consistent results.

The main disadvantages of this automated system are:

- expensive to set up in the first place and needs considerable testing
- always possible for a set of conditions to occur that were never considered during testing which could have safety implications (hence the need for a monitoring station)
- automated systems always need enhanced maintenance which can be expensive
- any computerised system is subject to cyberattacks no matter how good the system.

# Transport

As with industrial processes, many of the automated systems in transport refer to robotic systems (for example, autonomous buses/cars, autonomous trains and unpiloted aircraft).

But automated systems are still used in manually controlled transport, which includes cars, buses/lorries, trains and aircraft. (Examples 3 and 4 which follow, will use cars as the application.)

For example, modern trains will use an automatic signal control system. If the driver of the train goes through a red (stop) light, then the computer will automatically stop the train. This will make use of sensors at the side of the track sending signals to the on-board computer; actuators will be used to apply the brakes. Airplanes extensively use automatic pilots, which control the wing flaps, throttle and rudder to maintain the correct height, speed and direction.

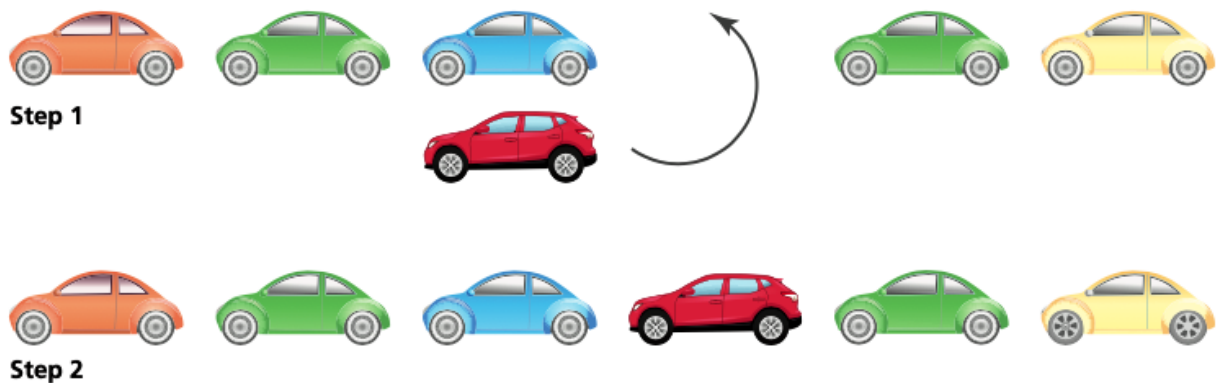


# Example 3

## Self-parking cars

The driver goes along the row of parked cars. On-board sensors and cameras gauge the size of any parking spaces, and the on-board computer warns the driver if a suitable space has been found. The driver then selects auto-parking and the on-board computer takes over. Actuators are used to operate the steering rack, brakes and throttle under the full control of the computer. This allows the car in figure to go from Step 1 to Step 2 automatically and complete the parking manoeuvre with no driver intervention.

Sensors in the bumpers of the car are both transmitters and receivers. The sensors transmit signals that bounce off objects and are reflected back. The car's on-board computer uses the amount of time it takes for the signal to return to the sensor to calculate the position of any objects. The sensors give the computer a 3D image of its surroundings. This allows the car to fit into its parking space automatically with no driver intervention. (Note: cheaper and older self-parking systems are not fully automatic; they require the driver to operate the brakes and throttle manually and only control the steering.)



The main advantages of this automated system are:

- allows the same number of cars to use fewer parking spaces
- avoids traffic disruption in cities (a manually controlled car takes several seconds to fit into a parking space)
- cars can fit into smaller spaces
- fewer dents and scratches to cars (reduced insurance claims)
- safer system since sensors monitor all objects, including young children (the car's manoeuvre will be stopped if any new object is encountered)
- very consistent results.

The main disadvantages of this automated system are:

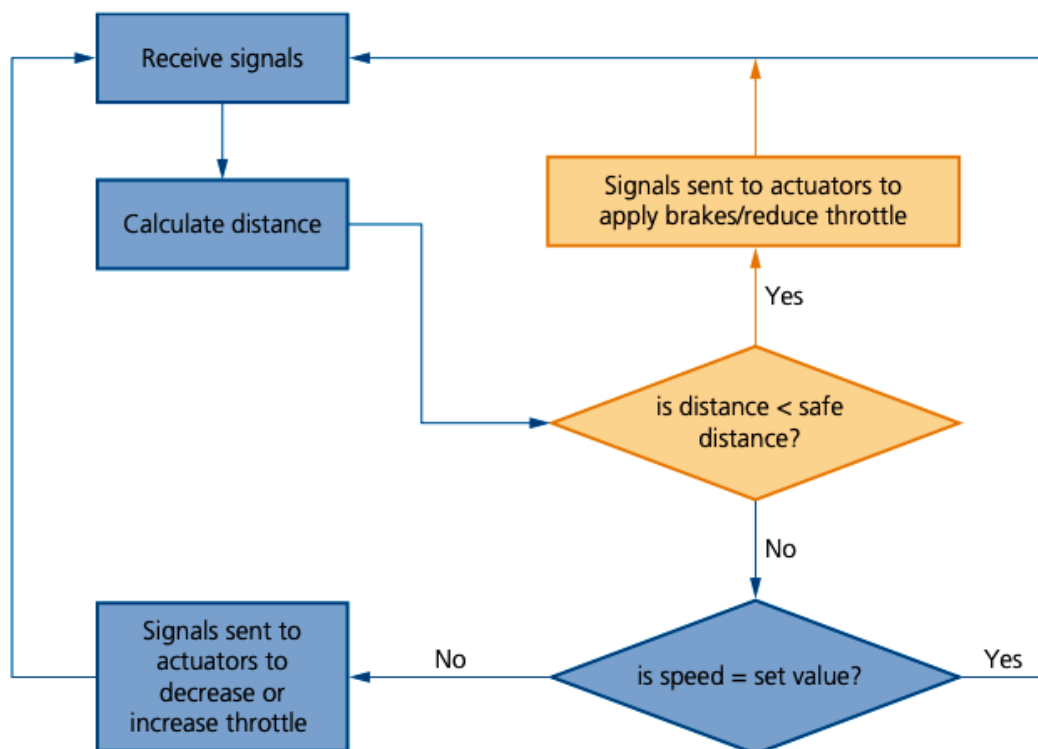
- over-reliance on automated systems by the driver (loss of skills)
- faulty/dirty sensors or cameras can send false data/images to the on-board computer which could lead to a malfunction
- kerbing of wheels is a common problem since the sensors may not pick-up low kerbs
- expensive option that doesn't really save the driver any money
- requires additional maintenance to ensure it functions correctly at all times.

# Example 4

## Adaptive cruise control

Adaptive cruise control makes use of sensors, an on-board computer and actuators to allow a car to remain a safe distance from another vehicle.

The driver will set a cruising speed (for example, 100 kph) on his touch screen in the car. Lasers (set into the bumpers of the car) are used to send out signals constantly. The lasers bounce off the vehicle in front of the car and are reflected back to the car's sensors. The time taken for the signal to bounce back is used by the on-board computer to calculate the distance between the two vehicles. If the car is getting too close to the vehicle in front, the computer will send signals to slow the car down. This is done by actuators applying the brakes and/or reducing the throttle. If the distance between vehicles is greater than the safe distance, the computer will check to see if the current speed equals the value set by the driver. If the speed is different to the set speed, the computer sends signals to the actuators to increase or decrease the throttle.



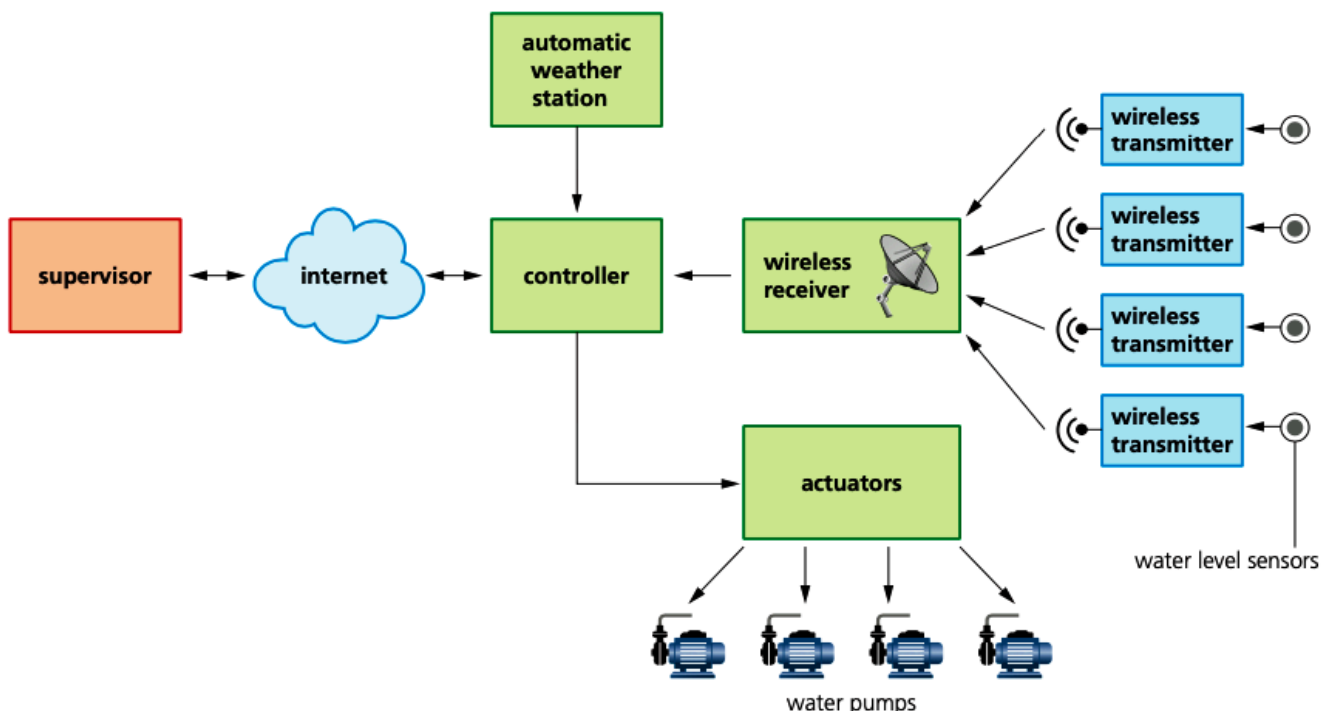
# Agriculture

There are many examples of the use of automated systems in agriculture. We will now consider one important example that is being used in Brazil to irrigate crops automatically.

## Example 5

### Automated system used in the Brazil irrigation system

The watering of crops (irrigation) is fully automatic and also involves considerable amounts of wireless transmission. This allows the system to be used in very remote areas that are vast in size – some of the fields are more than 10 km<sup>2</sup> in area.



Data from an automatic weather station (see next example) is received by the **controller** (a computer system) every ten minutes. This is particularly important if very wet or very dry conditions are being predicted or detected by the weather station. Ultrasonic water level sensors are used in the crop fields that measure the amount of water in the irrigation channels. The sensors send their data back via wireless transmitters. This data is then picked up by the wireless receiver, which sends the data back to the controller. The controller then uses this data, together with the data from the weather station, to decide whether it is necessary to stop or start a series of water pumps. This is done by sending signals to actuators, which operate the pumps. Although the whole system is fully automatic, a supervisor still monitors the process remotely. Using a schematic of a number of processes on a computer screen and via internet links to the controllers, the supervisor can oversee several irrigation processes from one central point. If the supervisor wishes to further increase or reduce the water supply in any of the irrigation systems, they can override the controller if necessary.

The main advantages of this automated system are:

- reduced labour costs since the system only needs a supervisor to monitor vast areas (if any maintenance is needed, then a dedicated team can cover all of the irrigation systems rather than having a separate team for each system)
- better and more efficient control of the irrigation process
- better control of precious resources, such as water
- faster response than a human having to manually check many kilometres of irrigation channels
- safer (temperatures in the fields could be 40°C and other risks could exist)
- different crops may require different irrigation requirements (for example, rice crops need flooding conditions, whereas orange trees like dry conditions); it is possible to program the controllers so that different growing conditions can be maintained simultaneously.

The main disadvantages of this automated system are:

- expensive to set up initially (expensive equipment needs to be bought)
- very high maintenance costs are associated with automated systems (also require specialist technicians if a fault occurs, which could be a problem in some remote areas of the world)
- increased need to maintain the water channels to ensure the system works correctly at all times (a blocked or collapsed channel wouldn't be picked up by the automated system, which could result in some areas being over-watered and some areas being starved of water).

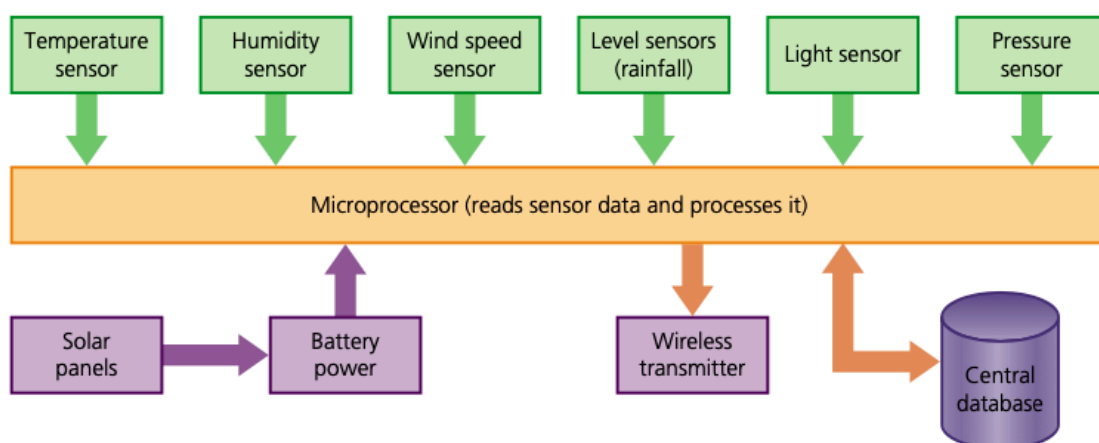
# Weather (stations)

Automated weather stations are designed to save labour and to gather information from remote regions or where constant weather data is a requirement. Automated weather stations require a microprocessor, storage (database), battery (usually with solar-powered charging) and a range of sensors: thermometer (to measure temperature)

- anemometer (to measure wind speed)
- hygrometer (to measure humidity)
- barometer (to measure air pressure)
- level sensor (to measure rain fall)
- light sensor (to measure hours of daylight).

The data from sensors is all sent to a microprocessor; any calculations are then done (for example, calculate hours of daylight, actual rainfall and wind direction). The data from the sensors and the calculated values are then stored on a central database. Some automated weather stations are sited near airports, where reports are sent out automatically every five minutes to pilots in the vicinity of the airport.

The only part of the weather station that needs to use actuators is the 'tipping bucket rain gauge'. At a pre-determined time interval, a signal is sent from the microprocessor to an actuator to operate a piston, which tips a bucket that was collecting rain water. The water is tipped into a vessel where level sensors are then used to measure the amount of rainfall that fell during the required time interval.



# Games

Gaming devices involve sensors to give a degree of realism to games:

accelerometers (these measure acceleration and deceleration and therefore measure and respond to tilting the gaming device forward/backward and side to side)

proximity sensors (used in smart touch pads; here electrodes are embedded in touch pads that can detect hand/finger position thus increasing user awareness).

Embedded accelerometers and proximity sensors (together with a microcontroller) in games consoles allow increased human interaction with the game. This allows players to take actions that simulate real events happening, giving a more immersive games experience.

# Lighting

Microprocessor-controlled lighting was discussed in Embedded systems. The example used in that chapter was the control of lighting in an office using:

- light sensors (to automatically switch lights on or off depending on the ambient lighting)
- motion sensors (to automatically turn lights on in a room when somebody enters)
- infrared sensors (to be used either as a motion detector or as part of the security system).



# Example 6

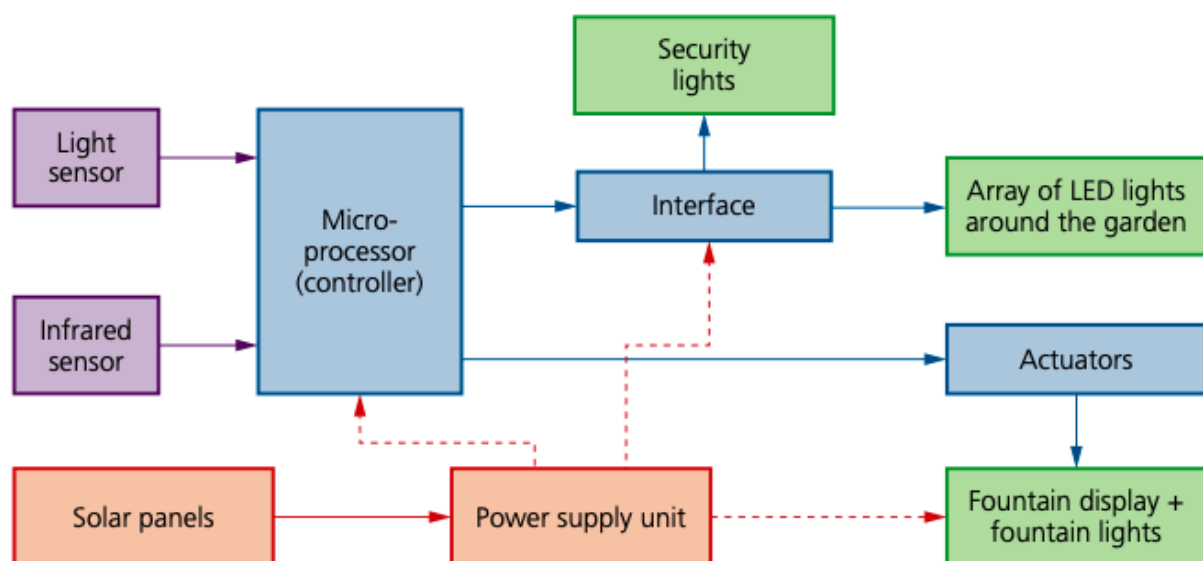
## Lighting system in a house

The example we will consider here is used in a house:

- where lights in the garden are turned on automatically when someone enters the garden or it turns dark
- where a lighting show is part of a microprocessor-controlled water fountain display; the lighting only comes on when it becomes dark.

As it becomes dark, the light sensor value will change, and the microprocessor will send signals to the interface to control the array of LED lights around the garden. Data from the infrared sensor would also be used (during day and night) as a security device whenever the house is unoccupied.

Also, as it becomes dark, the lighting show in the fountain could also be initiated. This could involve a pre-programmed display involving changing colours under the control of the microprocessor. The fountain display itself will also be under microprocessor control with signals being sent to actuators to turn water pumps on and off according to the installed program. The whole system will be fully automated.



The main advantages of this automated system are:

- it is possible to control light sources automatically
- a reduced energy consumption (since lights are only turned on when necessary)
- wireless connections can be chosen which are much safer (no trailing wires)
- longer bulb life (due to dimming or switching off when not in use)
- possible to program new light displays for various occasions.

The main disadvantages of this automated system are:

- expensive to set the system up in the first place
- if wireless connections chosen (for safety reasons), they can be less reliable than wired systems
- to ensure consistent performance, the automated system will require more maintenance (which can be expensive).

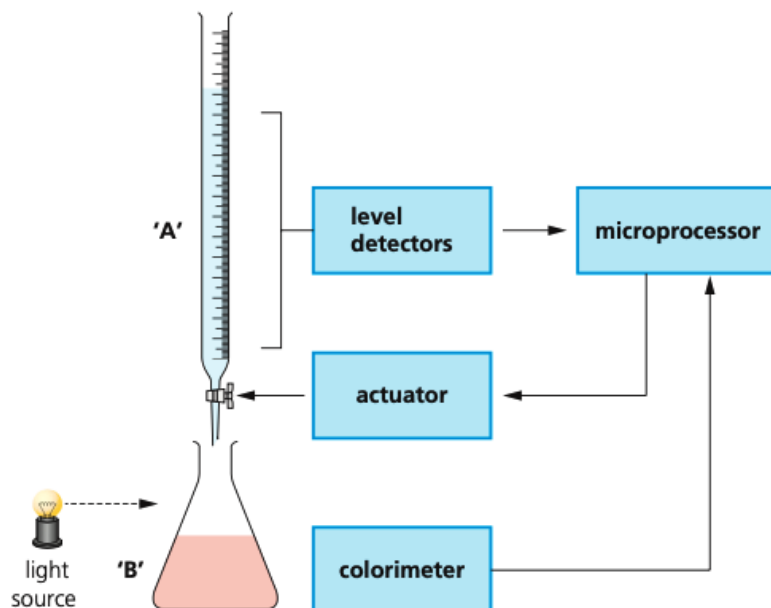
# Science

Automated systems in scientific research are widely used. There are literally thousands of possible applications. The example we will use here is the automatic control of a laboratory experiment which requires accuracy and repeatability.

## Example 7

### Chemical process in a laboratory

Imagine an experiment in a pharmaceutical laboratory where two chemicals are reacted together in a vessel. One of the chemicals is being added from a piece of equipment ('A') known as a **burette** (which has a tap to control the flow of liquid; the tap is operated automatically using a small actuator) to a reaction vessel ('B'). Once the reaction is complete, it turns a bright orange colour. The whole process is under microprocessor control:



The level sensors measure how much liquid is being added from 'A'; this data is sent to a microprocessor. Readings are also sent to the microprocessor from a colorimeter next to vessel 'B' (this instrument checks the colour of the chemical produced). The microprocessor controls the opening and closing of the tap in 'A'; this is done by sending signals to an actuator that operates the tap. This means the microprocessor has automatic control of the experiment with no human interaction needed. This type of automated system is of great benefit to pharmaceutical companies when developing new drugs and vaccines (several experiments can be carried out at the same time with only one person needed to monitor the system).

The main advantages of this automated system are:

- more consistent (repeatable) results
- less dangerous (especially if the chemicals used are hazardous)
- faster results (several different experiments can be done simultaneously)
- automatic analysis of the results is possible
- fewer highly trained staff needed for each experiment
- results/experiments can be monitored anywhere in the world in real time.

The main disadvantages of this automated system are:

- less flexible than when using human technicians
- security risks are always present if the data is being shared globally
- equipment can be expensive to buy and set up in the first place.

Finally, there are many automated systems being used in both industry and scientific research that incorporate artificial intelligence (AI). It is therefore worth considering the generic advantages of using AI in these automated systems:

- ability to access and store vast amounts of facts (very important in research)
- they are able to learn from huge amounts of available data that would overwhelm humans (or at the very least take them many months/years to do the same analysis)
- they are able to see patterns in results that could be missed by humans.

While all of this is positive, there are a few disadvantages in this approach:

- a change in skills set (is it the human or the AI that controls the research?)
- AI is dependent on the data which trains it.



# **Robotics**

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# What is robotics

The word robot comes from the Czech word robota (which means 'forced labour') and the term was first used in the 1920s play 'Rossum's Universal Robots'. The concept of the robot has fired the imagination of science fiction writers for countless years; indeed Isaac Asimov even composed his three laws of robotics:

- a robot may not injure a human through action or inaction
- a robot must obey orders given by humans, unless it comes into conflict with law 1
- a robot must protect itself, unless this conflicts with law 1.

So what is a robot in the real world? Robotics is a branch of (computer) science that brings together the design, construction and operation of robots. Robots can be found in:

- factories
  - welding parts together
  - spray-painting panels on a car
  - fitting windscreens to cars
  - cutting out metal parts to a high precision
  - bottling and labelling plants
  - warehouses (automatic location of items)



- in the home
  - autonomous floor sweepers
  - autonomous lawn mower
  - ironing robots (for example, 'dressman')
  - pool cleaning
  - automatic window cleaners
  - entertainment ('friend' robots)
- drones
  - unmanned aerial vehicles (UAVs) are drones that are either remotely controlled or totally autonomous using embedded systems
  - can be used in reconnaissance (for example, taking aerial photographs)
  - can be used to make parcel deliveries (for example, Amazon).

# Characteristics of a robot

To be correctly called a robot, they need to have the following characteristics:

1. Ability to sense their surroundings:

- this is done via sensors (such as light, pressure, temperature, acoustic, and so on)
- sensors allow a robot to recognise its immediate environment and gives it the ability to determine things like size, shape or weight of an object, detect if something is hot or cold, and so on; all sensor data is sent to a microprocessor or computer.

2. Have a degree of movement:

- they can make use of wheels, cogs, pistons, gears (etc.) to carry out functions such as turning, twisting, moving backwards/forwards, gripping or lifting
- they are mechanical structures made up of many parts (for example, motors, hydraulic pipes, actuators and circuit boards)
- they contain many electrical components to allow them to function
- can make use of end effectors (different attachments to allow them to carry out specific tasks such as welding, spraying, cutting or lifting).

3. Programmable:

- they have a 'brain' known as a controller that determines the action to be taken to perform a certain task (the controller relies on data sent from sensors or cameras, for example)
- controllers are programmable to allow the robots to do certain tasks.  
It is important to realise that robotics and artificial intelligence (AI) are almost two entirely different fields:

**TWO IMPORTANT NOTES:**

1. Many robots don't possess artificial intelligence (AI) since they tend to do repetitive tasks rather than requiring adaptive human characteristics.
2. It is important not to confuse physical robots with software robots such as:
  - search engine bots or WebCrawlers (these 'robots' roam the internet scanning websites, categorising them for search purposes)
  - chat bots (these are programs that pop up on websites that seem to enter some form of conversation with the web user)

According to our definition above, software robots are not true robots.

Physical robots can be classified as independent or dependent:

- Independent robots:
  - have no direct human control (they are said to be autonomous, for example, an autonomous vehicle)
  - can replace the human activity totally (no human interaction is required for the robot to function fully).
- Dependent robots:
  - have a human who is interfacing directly with the robot (the human interface may be a computer or a control panel)
  - can supplement, rather than totally replace, the human activity (for example, in a car assembly plant where both humans and robots work together to produce a car).

# The role of robots and their advantages and disadvantages

consider the use of robots in a number of areas, together with the advantages and disadvantages of using robots in each of these areas:

- industry
- transport
- agriculture
- medicine
- domestic (home) use
- entertainment.

## Industry

Robots are used in many areas of industry, from heavy lifting work right through to delicate procedures. Examples include: paint spraying of car bodies, welding bodywork on cars, manufacturing of microchips, manufacturing electrical goods and automatic warehouses.

Control of robots is either through embedded (built-in) microprocessors or directly linked to a computer system. Programming of the robot to do a series of tasks is generally done in two ways:

1. The robot is programmed with a sequence of instructions which allow it to carry out the series of tasks (for example, spraying a car body with paint).
2. Alternatively, a human operator manually carries out the series of tasks; this can be done in two ways. In our example, we will assume an object is being painted using a robot arm.



Figure above shows a robot arm equipped with a spray gun end-effector. Different end-effectors allow the robot arm to carry out many different tasks:

1. the robot arm is guided manually by a worker when spraying the object; each movement of the arm is stored as an instruction on the computer; or
2. the worker straps sensors to his own arm and sprays the object; each movement is stored as a set of instructions on a computer; the sensors send back information such as position relative to the object, arm rotation and so on – this information forms part of the instructions stored on the computer.

Whichever method is used, once the instructions have been saved, each series of tasks can then be carried out by a robot arm automatically. Each instruction will be carried out identically every time (for example, assembling parts in a television) giving a consistent product.

Robots are equipped with sensors so they can gather important information about their surroundings and also preventing them from doing 'stupid things'; for example, stopping a robot spraying a car if no car is present, or stop the spraying operation if the supply of paint has run out, and so on.

Robots are very good at repetitive tasks. However, if there are specialist tasks that require 'thinking' to cope with variable circumstances, (for example, making specialist glassware for some scientific work) then it is often better to still use human operators.

| Advantages                                                                                                | Disadvantages                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| robots are capable of working in conditions that may be hazardous to humans                               | robots can find it difficult to do 'non-standard' tasks (for example, windscreen being fitted to a car is cracked)                    |
| robots work 24/7 without the need to stop                                                                 |                                                                                                                                       |
| robots are less expensive in the long run (since there will be fewer salaries to pay)                     | robots can lead to higher unemployment amongst manual labour tasks                                                                    |
| robots are more productive than humans (higher productivity)                                              | there is a risk of deskilling when robots take over certain tasks (for example, welding and paint spraying)                           |
| although not necessarily more accurate, robots are more consistent                                        |                                                                                                                                       |
| robots are better suited to boring, repetitive tasks than humans (therefore less likely to make mistakes) | factories can now be moved to anywhere in the world where operation costs are lower (leading again to unemployment in some countries) |
| there will be less cost in heating and lighting (robots don't need good light or warmth)                  | robots are expensive to buy and set up in the first place                                                                             |

## Transport

Driverless vehicles are increasing in number every year. These are very complex robots, but the big problem is not really the technology (since problems will be ironed out through time), it is human perception. It will take a large leap of faith for humans to ride in a driverless car or an airplane with no pilot. We are already used to autonomous trains since these are used in many cities throughout the world. These systems have been generally accepted; but that is probably because trains don't overtake other trains and have a very specific track to follow

## Autonomous cars and buses

In this section, we will consider autonomous cars as our example. Autonomous cars use sensors, cameras, actuators and microprocessors (together with very complex algorithms) to carry out their actions safely. Sensors (radar and ultrasonics) and cameras allow the control systems in cars to perform critical functions by sensing the dynamic conditions on a road. They act as the 'eyes' and 'ears' of the car.

Microprocessors process the data received from cameras and sensors and send signals to actuators to perform physical actions, such as:

- change gear
- apply the brakes
- turn the steering wheel.

Cameras catch visual data from the surroundings, while radar and ultrasonics allow the vehicle to build up a 3D image of its surroundings (very important when visibility is poor, such as heavy rain, fog or at night). Suppose an autonomous car is approaching a set of traffic lights that are showing red. The first thing is the control system in the car needs to recognise the road sign and then check its database as to what action to take. Since the traffic light shows red, the microprocessor must send signals to actuators to apply brakes and put the gear into 'park'. Constant monitoring must take place until the light changes to green. When this happens, the microprocessor will again instruct actuators to put the car into first gear, release the brakes and operate the throttle (accelerator). This is a very complex set of operations since the microprocessor must constantly check all sensors and cameras to ensure moving off is safe (for example, has the car in front of it broken down or has a pedestrian started to cross the road, and so on).

| Advantages of autonomous vehicles                                                                                                                                                                                                       | Disadvantages of autonomous vehicles                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| safer since human error is removed leading to fewer accidents                                                                                                                                                                           | very expensive system to set up in the first place (high technology requirements)                                                                                                                                                                                                       |
| better for the environment since vehicles will operate more efficiently                                                                                                                                                                 | the ever-present fear of hacking into the vehicle's control system                                                                                                                                                                                                                      |
| reduced traffic congestion (humans cause 'stop-and-go' traffic known as ' <b>the phantom traffic jam</b> ', autonomous vehicles will be better at smoothing out traffic flow reducing congestion in cities)                             | security and safety issues (software glitches could be catastrophic; software updates would need to be carefully controlled to avoid potential disasters)                                                                                                                               |
| increased lane capacity (research shows autonomous vehicles will increase lane capacity by 100% and increase average speeds by 20%, due to better braking and acceleration responses together with optimized distance between vehicles) | the need to make sure the system is well-maintained at all times; cameras need to be kept clean so that they don't give false results; sensors could fail to function in heavy snowfall or blizzard conditions (radar or ultrasonic signals could be deflected by heavy snow particles) |
| reduced travel times (for the reasons above) therefore less commuting time                                                                                                                                                              | driver and passenger reluctance to use the new technology                                                                                                                                                                                                                               |
| stress-free parking for motorists (the car will find car parking on its own and then self-park)                                                                                                                                         | reduction in the need for taxis could lead to unemployment (imagine New York without its famous yellow cabs!)                                                                                                                                                                           |

## Autonomous trains

Autonomous (driverless) trains have been around for a number of years in a number of large cities. As with other autonomous vehicles, driverless trains make considerable use of sensors, cameras, actuators and on-board computers/microprocessors. Autonomous trains make use of a system called LiDaR (Light Detection and Ranging); LiDaR uses lasers which build up a 3D image of the surroundings. Other sensors (such as proximity sensors on train doors) and cameras (including infrared cameras) are all used for various purposes to help control the train and maintain safety. The control system in the train also makes use of global positioning satellite (GPS) technology, which allows accurate changes in speed and direction to be calculated. Again, actuators play a huge role here in controlling the train's speed, braking and the opening and closing of the train doors.



| Advantages of autonomous trains                                                                                                                             | Disadvantages of autonomous trains                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| this improves the punctuality of the trains                                                                                                                 | the ever-present fear of hacking into the vehicle's control system                                                                                                       |
| reduced running costs (fewer staff are required)                                                                                                            | system doesn't work well with very busy services (at the moment)                                                                                                         |
| improves safety since human error is removed                                                                                                                | high capital costs and operational costs initially (that is, buying the trains, expensive signalling and control equipment and the need to train staff)                  |
| minimises energy consumption since there is better control of speed and an optimum service requires less energy (trains stuck in stations still use energy) | ensuring passenger behaviour is acceptable particularly during busy times (for example, jamming doors open on trains, standing too near the edge of platforms and so on) |
| it is possible to increase the frequency of trains (automated systems allow for shorter times between trains)                                               | passenger reluctance to use the new technology                                                                                                                           |
| it is easier to change train scheduling (for example, more trains during busier times)                                                                      | no drivers mean there will be a need for CCTV to monitor railway stations                                                                                                |

## Autonomous (un-piloted) airplanes

Airplanes have used auto-pilots for many years to control flights. Human pilots only take over during take-off and landing. Autonomous (pilotless) airplanes would make even more extensive use of sensors, actuators and microprocessors to control all stages of the flight. Some of the main features of a control system on a pilotless airplane would include:

- sensors to detect turbulence to ensure smooth flights
- an increase in self-testing of all circuits and systems
- sensors that would automatically detect depressurisation in the cabin; thus allowing for quick stabilisation of the airplane
- use of GPS for navigation and speed calculations
- use of actuators to control, for example, throttle, flaps (on the wings) and the rudder.

| Advantages of pilotless airplanes                                                                                              | Disadvantages of pilotless airplanes                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| improvement in passenger comfort (reasons given earlier)                                                                       | security aspects if no pilots on-board (for example, handling terrorist attacks)                                                      |
| reduced running costs (fewer staff are required)                                                                               | emergency situations during the flight may be difficult to deal with                                                                  |
| improved safety (most crashes of airplanes have been attributed to pilot-induced errors)                                       | hacking into the system (it might be possible to access flight controls)                                                              |
| improved aerodynamics at the front of the airplane since there would no longer be the need to include a cockpit for the pilots | passenger reluctance to use the new technology                                                                                        |
|                                                                                                                                | software glitches (recent software issues with modern airplanes have highlighted that software glitches can have devastating results) |

## Agriculture

With the world's population predicted to reach nine billion by the year 2050, more efficient agriculture via increased use of robotics is inevitable. Robots could replace slow, repetitive and dull tasks allowing farmers to concentrate on improving production yields. We will consider the following five areas where robotics could play a big role:

- harvesting/picking of vegetables and fruit
- weed control
- phenotyping (plant growth and health)
- seed-planting and fertiliser distribution
- autonomous labour-saving devices.

## **Harvesting and picking**

- robots have been designed to do this labour-intensive work; they are more accurate (only pick ripe fruit, for example) and much faster at harvesting
- for the reasons above, this leads to higher yields and reduces waste (for example, vegebot (Cambridge University) uses cameras to scan, for example, a lettuce and decide whether or not it is ready to be harvested
- a second camera in vegebot (near the cutting blades) guides an arm to remove the lettuce from its stalk with no damage.

## **Weed control**

- weed management robots can distinguish between a weed and crop using AI
- examples of weed control robots are being used in France (by Mouton- Rothschild) to remove weeds between grape vines in their vineyards; this saves considerably on labour costs and improves vine growth
- weed control robots use GPS tracking to stay on course to move along the rows of vines and remove the weeds; a weed removal blade is operated by an actuator under the control of the controller (microprocessor) in the robot
- very often a drone (flying robot) is used first to do an aerial view of the vineyard, so that a programmed course of action can be produced, which is then sent to the weed control robot's memory.

## Phenotyping

- phenotyping is the process of observing physical characteristics of a plant in order to assess its health and growth
- robots designed to do phenotyping are equipped with sensors (including spectral sensors and thermal cameras) that can create a 3D image/model of the plant, thus allowing it to be monitored for health and growth
- machine learning (see Section 6.3) is used to recognise any issues with leaves (for example, if they have a blight or have the wrong colour) so that the robot can convey this back to the farmer
- these robots are much more accurate and faster at predicting problems than when done manually.

## Seed-planting drones and fertiliser distribution

- drones (flying robots) can produce an aerial image of a farm sending back a 'bird's eye view' of the crops and land
- they allow seed-planting to be done far more accurately
- they also allow for more efficient fertiliser-spreading to reduce waste and improve coverage (this is much more efficient than conventional crop spraying)
- drones can also be used in cloud seeding where the drone can add silver iodide crystals to a cloud forcing it to give up its rainwater
- the drones use a very complex camera system to target seeding and allow fertiliser spraying.

## **Autonomous agriculture devices**

Several of the devices described above could be referred to as autonomous. The following list summarises some of the devices that can work independently of humans:

- grass mowers/cutters
- weeding, pruning and harvesting robots
- seeding robots
- fertiliser spraying
- all of these devices use sensors and cameras to go around obstacles, or they can even be programmed to 'go to sleep' if the weather turns bad.

# Medicine

- robots are used in surgical procedures, which makes the operation safer and also makes the procedures quicker and less costly
- robots can be used from monitoring patients to doing actual minor surgery
- the disinfecting of rooms and operating theatres can all be done by autonomous robots (similar to the types described in agriculture)
- robots can take blood samples from patients:
  - less painful to patients since the robot is better at determining a 'good vein'
  - safer to doctors and nurses if the patient has an infectious disease
  - doctors and nurses can be freed up to do more skilled work
- microbots can be used in target therapy:
  - these use microscopic mechanical components (including microprocessor) to localise a drug or other therapy to target a specific site causing less damage to surrounding tissue
- prosthetic limbs are now mini robots in their own right (since they meet the three characterisations of what defines a robot)
  - bionic skins and neural implants that interface with the human nervous system (of the damaged limb) giving feedback to allow for better control of the prosthetic limb (again sensors and actuators are used to give human-like responses, such as grip).

# Domestic robots

Robots used around the house vary from devices to carry out household chores through to devices used to entertain people. For example:

- autonomous vacuum cleaners:
  - these use proximity sensors and cameras to avoid bumping into obstacles and allows them to cover a whole room automatically
  - these robots have a microprocessor to control the overall operation of the device; this also allows the user to program the device
  - actuators are used to control motors which allow movement forward/ backward and from side to side
- autonomous grass cutters (mowers):
  - these use the same type of sensor, camera, microprocessor and actuator set up as vacuum cleaners
- personal assistants (such as 'Vector')
  - this is a robot controlled by a micro-processor that also uses cloud connectivity to connect to the internet
  - it understands voice commands (using a microphone) and will answer any questions it is asked
  - it also makes use of an HD camera, utilising computer vision, allowing it to recognise somebody's face as well as navigate a room (using proximity sensors and actuators) to steer around objects in its way.

# Robots used in entertainment

The use of robots in the entertainment industry is increasing. They are now found in areas such as:

- entertainment parks and arenas/venues
- the film and TV industry.

The following examples indicate where robots are being used in the world of entertainment. The reader is advised to research the ever-increasing number of examples.

- theme parks are now using autonomous robots to entertain visitors to the park; these robots (often dressed as cartoon characters) can interact with visitors to allow them to engage safely with the theme park attractions and make the whole experience 'more realistic'
- music festivals are much more immersive for the audience; robotic methods are used to control lighting (including laser displays), visual effects and animation (e.g. superimposing an actor's image onto a robotic caricature and synchronising mouth movements); the visual performances can be fully synchronised with the music
- use of robots to control cameras; for example, keeping them steady and auto- focusing when moving around a scene; the movie *Gravity* used many robots to operate cameras, props and the actors (for example, to give an actor the appearance of moving around in the vacuum of space uncontrollably, robot arms were used to simulate human behaviour and produce life-like moving images)
- humanoid robots (either remote-controlled or pre-programmed) can perform 'stunt' action in movies/ television by performing tasks impossible for a human to do; they use CGI (computer-generated imagery) and image capture techniques to generate special effects
- robots are capable of producing special effects with a precision, speed and coordination which is beyond human capabilities; actions and special effects can be synchronised to within a millisecond and produce fully coordinated/ synchronised sound effects (e.g. movement of the mouth to match the sounds produced in a realistic manner).



# **Artificial intelligence (AI)**

## **Automated and emerging technologies**

Zain Merchant

Artificial intelligence (AI) is a branch of computer science dealing with the simulation of intelligent human behaviour by a computer. This is often referred to as the cognitive functions of the human brain (that is, the mental process of acquiring knowledge and understanding through thought, experience and the five senses). All of these cognitive functions can be replicated in a machine, and they can be measured against human benchmarks such as reasoning, speech and sight.

# Characteristics of AI

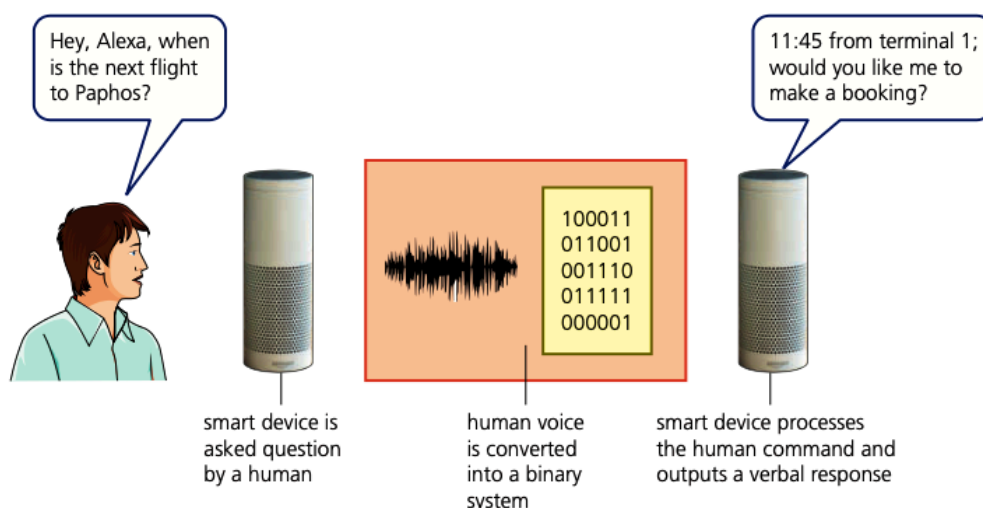
Essentially, AI is really just a collection of rules and data, and the ability to reason, learn and adapt to external stimuli. AI can be split into three categories:

- narrow AI – this occurs when a machine has superior performance to a human when doing one specific task
- general AI – this occurs when a machine is similar (not superior) in its performance to a human doing a specific task
- strong AI – this occurs when a machine has superior performance to a human in many tasks.

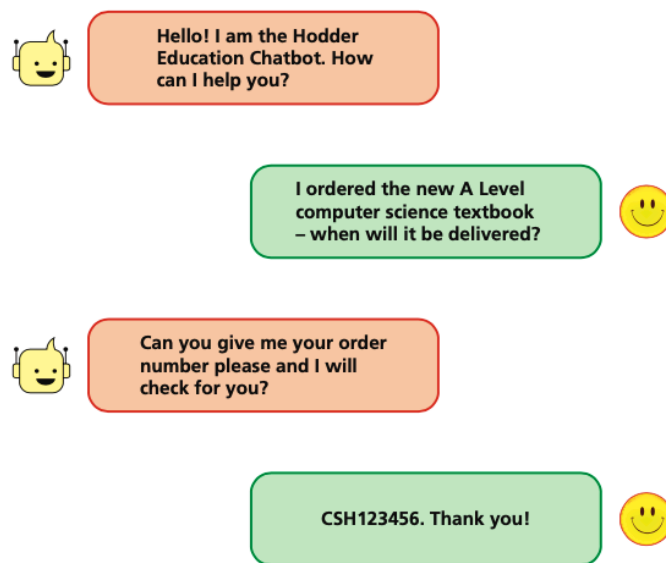
Reasoning is the ability to draw reasoned conclusions based on given data/ situations. Deductive reasoning is where a number of correct facts are built up to form a set of rules which can then be applied to other problems (for example, if AI is used to produce the perfect cup of tea based on a number of facts, the machine will learn from the experience and apply its new rules in the making of a cup of coffee, hot chocolate, and so on – modifying its methodology where necessary). By carrying out a sequence of steps, the AI machine can learn, and next time it will know how to do the task more effectively and even apply it to a novel/new situation. Thus the AI system is capable of learning and adapting to its surroundings. AI can very quickly discern patterns (which in some cases, humans cannot) and then make predictions by adapting to the new data

Examples of AI include:

- news generation based on live news feeds
- smart home devices (such as Amazon Alexa, Google Now, Apple Siri and Microsoft Cortana):
  - the AI device interacts with a human by recognising verbal commands
  - it learns from its environment and the data it receives
  - the device becomes increasingly sophisticated in its responses, thus showing the ability to use automated repetitive learning



- use of chatbots that interact through instant messaging, artificially replicating patterns of human interactions using AI to respond to typed or voice messages; when a question is asked, the chatbot responds using the information known at the time:



- use of chatbots that interact through instant messaging, artificially replicating patterns of human interactions using AI to respond to typed or voice messages; when a question is asked, the chatbot responds using the information known at the time:
- autonomous cars
- facial expression recognition
  - algorithms identify key facial landmarks such as the corners of the eyebrows, corners of the mouth, and so on
  - a combination of these landmarks can be used to map emotions (such as anger, fear, joy and surprise).

# AI systems

- expert system – a computer system that mimics the decision-making ability of a human; expert systems use AI to simulate the judgement and behaviour of a human or organisation that has expert knowledge and experience
- machine learning – this is the science of training computers with sample data so that they can go on to make predictions about new unseen data, without the need to specifically program them for the new data.

## Expert systems

Expert systems are a form of AI that has been developed to mimic human knowledge and experiences. They use knowledge and inference to solve problems or answer questions that would normally require a human expert.

For example, suppose the user was investigating a series of symptoms in a patient. The expert system would ask a series of questions, and the answers would lead to its diagnosis. The expert system would explain its reasoning with a statement such as 'impaired vision, lack of coordination, weak muscles, slurred speech and the patient used to work in a paint factory – the diagnosis is mercury poisoning' – the user could then probe deeper if necessary.

The expert system will supply a conclusion and any suggested actions to take and it will also give the percentage probability of the accuracy of its conclusions (for example, the following statement could be made 'Based on the information given to me, the probability of finding oil bearing rocks in location 123AD21G is about 21%').

There are many applications that use expert systems:

- oil and mineral prospecting
- diagnosis of a patient's illness
- fault diagnostics in mechanical and electronic equipment
- tax and financial calculations
- strategy games, such as chess
- logistics (efficient routing of parcel deliveries)
- identification of plants, animals and chemical/biological compounds.

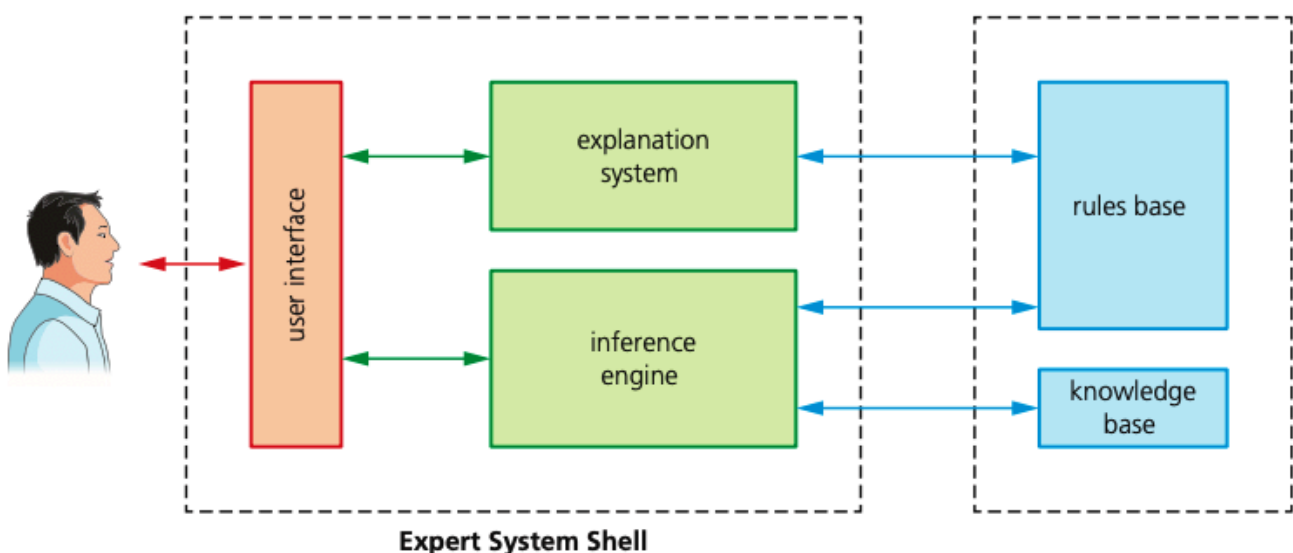
Expert systems have many advantages:

- they offer a high level of expertise
- they offer high accuracy
- the results are consistent
- they have the ability to store vast amounts of ideas and facts
- they can make traceable logical solutions and diagnostics
- it is possible for an expert system to have multiple expertise
- they have very fast response times (much quicker than a human expert)
- they provide unbiased reporting and analysis of the facts
- they indicate the probability of any suggested solution being correct.

Expert systems also have disadvantages:

- users of the expert system need considerable training in its use to ensure the system is being used correctly
- the set up and maintenance costs are very high
- they tend to give very 'cold' responses that may not be appropriate in certain medical situations
- they are only as good as the information/facts entered into the system
- users sometimes make the very dangerous assumption that they are infallible.

So what makes up an expert system? Figure shows the typical structure of an expert system:



## **User interface**

- method by which the expert system interacts with a user
- interaction can be through dialogue boxes, command prompts or other input methods
- the questions being asked usually only have Yes/No answers and are based on the responses to previous questions.

## **Inference engine**

- this is the main processing element of the expert system
- the inference engine acts like a search engine examining the knowledge base for information/data that matches the queries
- it is responsible for gathering information from the user by asking a series of questions and applying responses where necessary; each question being asked is based on the previous responses
- the inference engine is the problem-solving part of the expert system that makes use of inference rules in the rules base
- since the knowledge base is a collection of objects and attributes, the inference engine attempts to use information gathered from the user to find an object that matches (making use of the rules base to find a match)

## **Knowledge base**

- the knowledge base is a repository of facts
- it stores all the knowledge about an area of expertise obtained from a number of expert resources
- it is basically a collection of objects and their attributes; for example:



| Object | Attribute 1 | Attribute 2  | Attribute 3    | Attribute 4        | Attribute 5              | Attribute 6         |
|--------|-------------|--------------|----------------|--------------------|--------------------------|---------------------|
| dog    | mammal      | can be a pet | lives on land  | makes bark sounds  | body is covered in fur   | walks on 4 legs     |
| whale  | mammal      | not a pet    | lives in water | makes sonic sound  | body covered in skin     | swims; no legs      |
| duck   | bird        | not a pet    | lives in water | makes quack sounds | body covered in feathers | swims; has two legs |

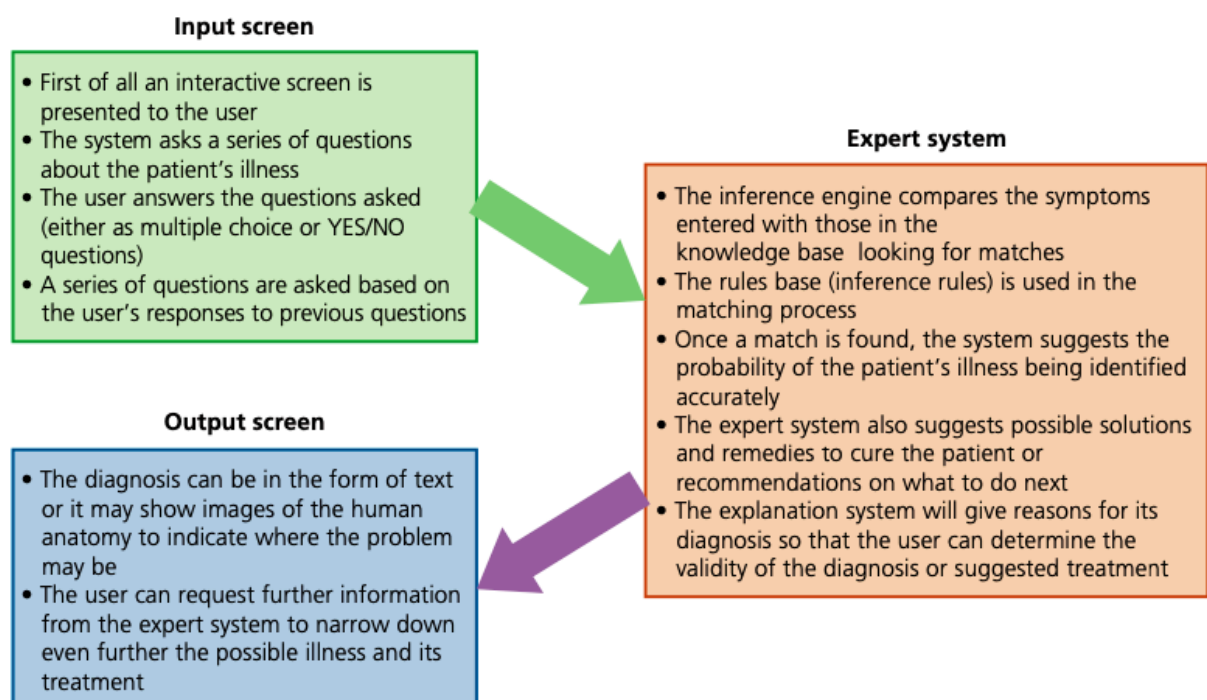
- so if we had a series of questions:
  - is it a mammal? YES
  - can it be a pet? NO
  - does it live in water? YES
  - does it make sonic sounds? YES
  - is its body covered in skin? YES
  - does it have any legs? NO conclusion: it is a WHALE.

### Rules base

- the rules base is a set of inference rules
- inference rules are used by the inference engine to draw conclusions (the methods used closely follow human reasoning)
- they follow logical thinking like the example above; usually involving a series of 'IF' statements, for example:

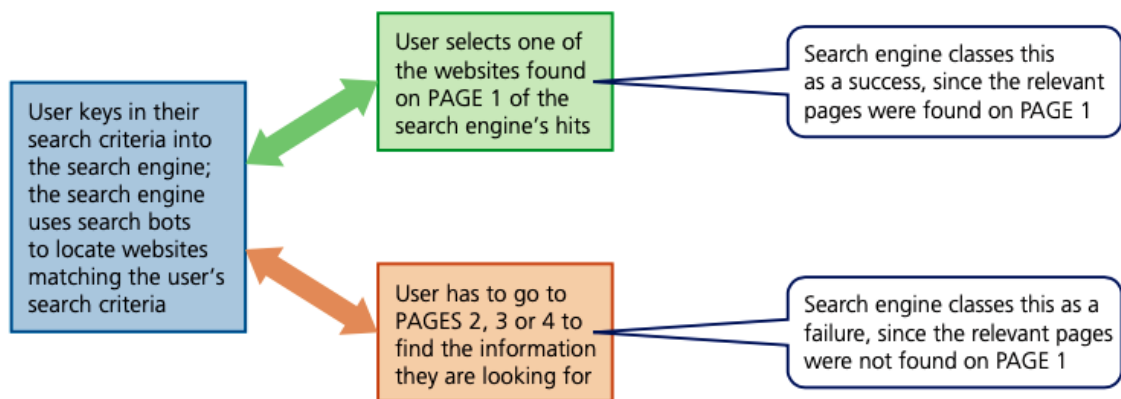
**IF continent = "South America" AND language = "Portuguese" THEN country = "Brazil"**

- Setting up an expert system
- information needs to be gathered from human experts or from written sources such as textbooks, research papers or the internet
- information gathered is used to populate the knowledge base that needs to be first created
- a rules base needs to be created; this is made up of a series of inference rules so that the inference engine can draw conclusions
- the inference engine itself needs to be set up; it is a complex system since it is the main processing element making reasoned conclusions from data in the knowledge base
- the user interface needs to be developed to allow the user and the expert system to communicate
- once the system is set up, it needs to be fully tested; this is done by running the system with known outcomes so that results can be compared and any changes to the expert system made.



# Machine learning

Machine learning is a sub-set of artificial intelligence (AI), in which algorithms are 'trained' and learn from their past experiences and examples. It is possible for the system to make predictions or even take decisions based on previous scenarios. They can offer fast and accurate outcomes due to very powerful processing capability. One of the key factors is the ability to manage and analyse considerable volumes of complex data; some of the tasks would take humans years to complete without the help of machine learning techniques. One example that uses machine learning are the most sophisticated search engines:



The search engine will learn from its past performance, meaning its ability to carry out searches becomes more and more sophisticated and accurate.

## Differences between AI and machine learning

| AI                                                                    | Machine learning                                                                                     |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| represents simulated intelligence in machines                         | this is the practice of getting machines to make decisions without being programmed to do so         |
| the aim is to build machines that are capable of thinking like humans | the aim is to make machines that learn through data acquisition, so that they can solve new problems |

# Example 1

## Categorizing email as spam

Consider email messages such as '*You have won \$2 million in the National Lottery*'; how can machine learning determine that this email should be put into your spam folder?

- A machine learning algorithm collects data about emails, such as email content, headers, senders name/email address and so on.
- It carries out a 'cleaning' process by removing **stop words** (for example, the, and, a) and punctuation, leaving only the relevant data.
- Certain words/phrases are frequently used in spam (for example, lottery, earn, full-refund) and indicate that the incoming email is very likely to be spam.
- The machine learning model is built and a 'training data set' is used to train the model and make it learn using past email known to be spam.
- Once it is evaluated, the model is fine-tuned and tested live.

# Example 2

## Detection of fraudulent activity

- Data is gathered by a survey or web scraping; for example, to detect credit card fraud, information about customers is gathered, such as types of transactions (does the customer buy designer clothes?), shopping habits and certain personal data.
- Redundant data is then removed; this needs to be carefully done to remove the possibility of wrong predictions.
- The most important machine learning step: the algorithm is trained through real examples of customer purchasing behaviour.
- A model is built based on learning from the training data, and the machine learning algorithm can now be used to detect fraud (for example, if a customer spends an unusual amount on a piece of jewellery, there is a high chance a fraudulent activity has taken place).
- The machine learning model is then fully tested with known data and known outcomes; the system is modified if it hasn't met its criteria to detect fraudulent activity.

